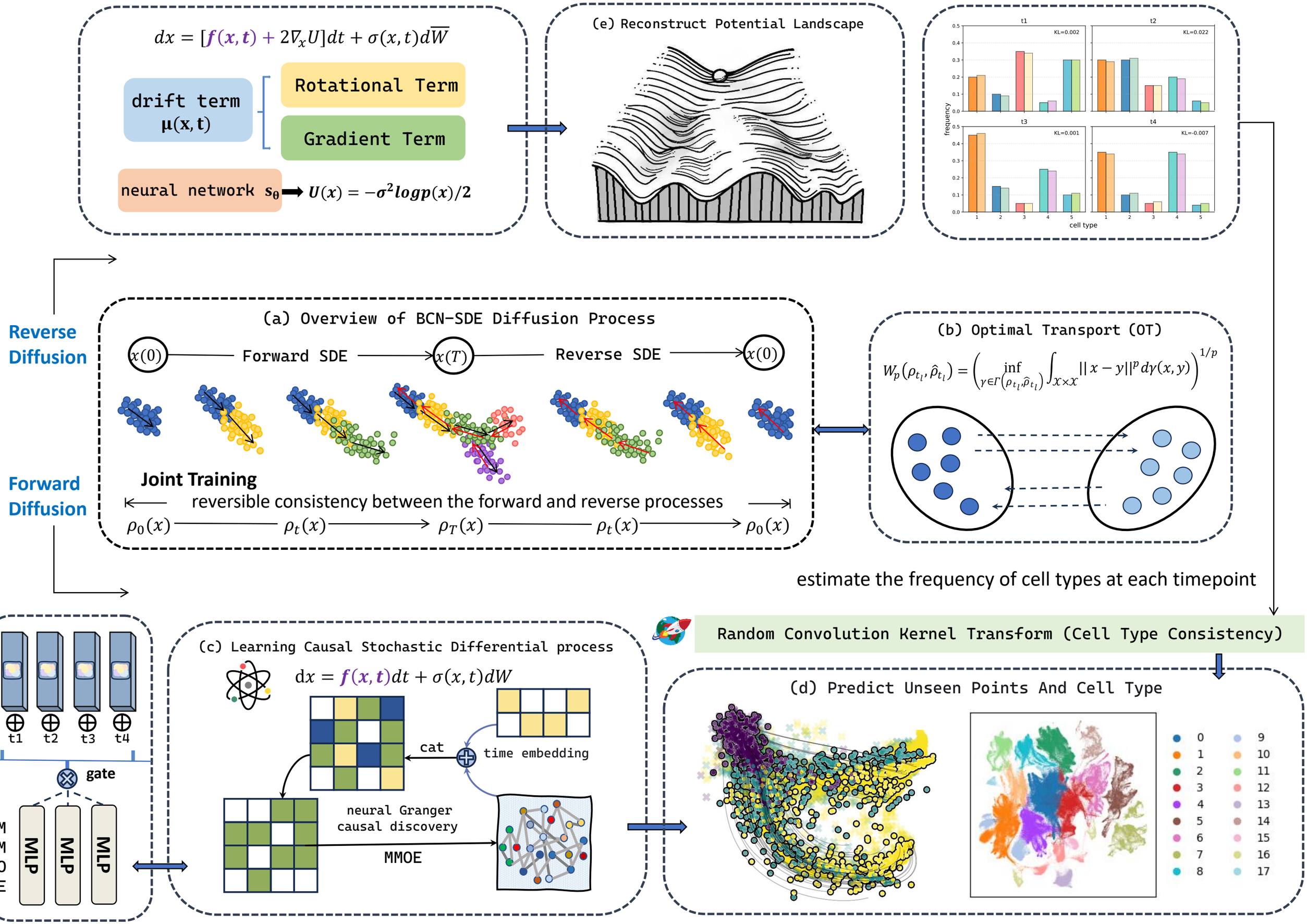
$dx = [f(x,t) + 2\nabla_x U]dt + \sigma(x,t)d\overline{W}$
drift term $\mu(x,t)$
Rotational Term
Gradient Term
neural network s_θ
$U(x) = -\sigma^2 logp(x)/2$
(e) Reconstruct Potential Landscape
t1
t2
t3
t4
KL=0.002
KL=0.022
KL=0.001
KL=-0.007
frequency
cell type
Reverse Diffusion
Forward Diffusion
(a) Overview of BCN-SDE Diffusion Process
$x(0)$
Forward SDE
$x(T)$
Reverse SDE
$x(0)$
Joint Training
reversible consistency between the forward and reverse processes
$\rho_0(x)$
$\rho_t(x)$
$\rho_T(x)$
$\rho_t(x)$
$\rho_0(x)$
(b) Optimal Transport (OT)
$W_p(\rho_{t_l}, \hat{\rho}_{t_l}) = \left(\inf_{\gamma \in \Gamma(\rho_{t_l}, \hat{\rho}_{t_l})} \int_{\mathcal{X}\times\mathcal{X}} ||x-y||^p d\gamma(x,y)\right)^{1/p}$
estimate the frequency of cell types at each timepoint
Random Convolution Kernel Transform (Cell Type Consistency)
t1
t2
t3
t4
gate
MMOE
MLP
MLP
MLP
(c) Learning Causal Stochastic Differential process
$dx = f(x,t)dt + \sigma(x,t)dW$
cat
time embedding
neural Granger causal discovery
MMOE
(d) Predict Unseen Points And Cell Type
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17